

Supplementary Material for “Information-Set Conditional Integrity Auditing for Hybrid Quantum-Classical Kernel Workflows”

Roberto Fernández-Barrios, Iker Pastor-López, Amaia Pikatza-Huerga, and Pablo García Bringas
Faculty of Engineering, University of Deusto, Bilbao, Spain



1 PURPOSE AND CLAIM BOUNDARY

This supplement records the complete experimental design, coverage contract, validation checks, and reproduction mapping behind the main article. It adds no QPU, calibrated-noise, scheduling, provider-security, multi-tenancy, or operational Fleet Management evidence. The exact claims remain bounded to data-to-evaluation interventions, ideal-statevector quantum kernels, controlled binomial shot emulation, and a local executable contract.

2 FROZEN EXPERIMENTAL DESIGN

2.1 Expansion Environments

The expansion comprises eight fixed environments (Table 1). “OOD” means a prespecified source-to-target construction, not a sampled deployment population. Every environment uses five split seeds and two nested model seeds. All evaluate projected dimensions 8, 10, and 12. The CICIDS scale gate uses 256/256 train/evaluation samples; all others use 128/128. The two additional ID datasets retain all empirically distinct quantum profiles. Scale and temporal gates compare SVC-RBF with QSVC-ZZ.

Each staged table contains 3,000 balanced binary rows before experimental subsampling. Numeric non-finite values are replaced during staging; categorical fields are excluded. These choices produce a controlled integrity audit but do not reproduce operational prevalence. Several temporal targets yield clean performance near chance, limiting the impact that later interventions can express; the exact information-set invariances do not depend on clean headroom.

2.2 Perturbation Suite

The paper-core suite crosses six mechanisms with nominal strengths 0.02, 0.05, and 0.10 (Table 2). Artifact identifiers retain the historical family string `target_shift`; the scientific text uses *evaluation-label intervention* because these rows do not instantiate the statistical label-shift problem.

The feature mechanisms may alter representation, scores, predictions, and the reported conclusion. Both label mechanisms keep the feature representation, scores, and predictions fixed. Prior-preserving flips also retain the empirical class

histogram. Equal mechanism weighting is a deliberately transparent stress profile, not an estimate of real incident frequency.

3 INFORMATION-SET COVERAGE CONTRACT

Table 3 records what each evidence regime can establish in the implemented study. Structural blindness means that the sensor’s complete input is unchanged by construction; it is stronger and narrower than observing zero empirical response in a finite sample.

Every machine-readable coverage record contains seven fields:

- 1) workflow boundary;
- 2) protected asset;
- 3) adversary or failure capability, including fixed elements;
- 4) available information regime;
- 5) sensor, reference, threshold, and false-alarm budget;
- 6) covered perturbations and residual blind region; and
- 7) response or countermeasure.

This record prevents a sensor response from being promoted into a global security claim.

4 DEDUPLICATION AND STATISTICAL UNITS

Gate 1 emits 7,980 raw model-intervention rows. Different quantum-map jobs repeat the same classical SVC pipeline, producing 3,420 repeated rows. The builder compares every repeated key across performance, impact, and raw signal columns before retaining one representative; 4,560 unique observations remain.

The expansion emits 13,680 raw rows from 360 completed CSV/JSON job pairs. Its multi-map ID gates contain 2,280 repeated SVC rows, again accepted as duplicates only after numerical identity checks; 11,400 unique observations remain. The strict builder fails if a job is missing, an environment contains an unexpected model/dimension/seed/attack design, or a duplicated row disagrees.

Coverage counts use prespecified cells as their denominators. For the secondary suite-average model profile, nested

TABLE 1
Prespecified expansion environments.

ID	Environment	Training source	Evaluation source	Samples	Profiles
E1	CICIDS scale/ID	Frozen CICIDS subset	Same staged source	256/256	SVC, ZZ
E2	UNSW-NB15 ID	Balanced staged UNSW	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ- equivalent
E3	ToN-IoT ID	Balanced staged ToN-IoT	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ- equivalent
E4	CICIDS Tue→Wed	Tuesday traffic	Wednesday traffic	128/128	SVC, ZZ
E5	CICIDS Tue→Fri-PortScan	Tuesday traffic	Friday PortScan traffic	128/128	SVC, ZZ
E6	CICIDS Wed→Thu-Web	Wednesday traffic	Thursday Web traffic	128/128	SVC, ZZ
E7	CICIDS Wed→Fri-AM	Wednesday traffic	Friday morning traffic	128/128	SVC, ZZ
E8	UNSW temporal OOD	Prespecified UNSW source period	Prespecified UNSW target period	128/128	SVC, ZZ

TABLE 2
Frozen 18-condition intervention suite.

IDs	Mechanism	Strengths
1-3	Feature sign flip	0.02, 0.05, 0.10
4-6	Feature-wise mean shift	0.02, 0.05, 0.10
7-9	Scaling drift	0.02, 0.05, 0.10
10-12	Feature dropout	0.02, 0.05, 0.10
13-15	Prior-preserving evaluation-label flip	0.02, 0.05, 0.10
16-18	Random evaluation-label flip	0.02, 0.05, 0.10

model-seed values are averaged inside each of five split seeds. The split seed is the inferential unit, and the interval is

$$\bar{d} \pm t_{0.975,4} \frac{s_d}{\sqrt{5}}, \quad (1)$$

where d is the within-split ZZ-minus-SVC impact difference. The intervals are not multiplicity-adjusted and do not support population transport. Results over 20 split-model combinations are sensitivity analyses only.

5 EXACT-STATEVECTOR ENGINE VALIDATION

The expansion evaluator replaces pairwise compute-uncompute execution with a mathematically equivalent ideal-statevector calculation. It prepares each unique state $|\phi(x)\rangle$ once and forms

$$K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2 \quad (2)$$

by matrix multiplication. Symmetric blocks use the same eigenvalue-clipping PSD repair as the frozen Qiskit reference.

Validation has four layers:

- 1) symmetric and cross-kernel unit matrices agree with the reference at relative and absolute tolerance 10^{-10} ;
- 2) symmetry, unit diagonal, numerical range, cache bounds, and cache reuse are unit-tested;
- 3) a CICIDS 128/128 end-to-end replay (split/model seed 42, dimension 8, ZZ, paper-core suite) preserves every categorical, discrete performance, impact, and integrity-signal endpoint; and
- 4) the prespecified 256/256 pilot completes.

The two end-to-end outputs each have 38 rows and 126 columns. Only ROC-AUC and its derived drops differ,

with maximum absolute difference 1.221896×10^{-4} , approximately one half-tie increment for 64 positive and 64 negative evaluation examples. This supports an explicit 2×10^{-4} ROC-AUC tolerance, not bitwise identity of continuous scores. The engine is tagged `exact_statevector`; it supplies no finite-shot, noise-model, or hardware evidence.

6 QUANTUM GATE: CONDITIONS AND COVERAGE

The quantum gate crosses five split seeds, dimensions 4/6/8, and 11 conditions, giving 165 cells. Table 4 reports response fractions over the 15 split-dimension cells per condition. A value of one means nonzero raw response in all 15 cells at the frozen tolerance; it is not calibrated power.

All nine frozen acceptance checks pass: 165 cells are present; clean evidence is invariant; benign transpilation and common-unitary rewrite alter provenance but preserve semantic kernels and outputs; every data-dependent circuit mutation changes the semantic kernel; symmetry and diagonal sensors catch their respective malformed edits; the PSD-preserving mixture passes algebra but is exposed by trusted provenance/semantics; and both shot estimators produce nonzero repeat discrepancy in every cell.

7 EXECUTABLE CONTRACT DETAILS

The prototype evaluates four ordered contracts:

- 1) **Input/preprocessing**: input-array and preprocessing-declaration hashes; mismatch blocks.
- 2) **Circuit/kernel**: circuit/kernel provenance, semantic probes, symmetry, diagonal, eigenvalue, and PSD-repair evidence; an equivalent rewrite passes only if explicitly approved.
- 3) **Execution/result**: declared execution metadata, repeated- estimation discrepancy, and prediction provenance; approved stochastic execution is held for adjudication.
- 4) **Evaluation/report**: item-level labels are bound to predictions and balanced accuracy is recomputed; prior-preserving corruption blocks even when the marginal is identical.

Contracts form an ordered SHA-256 chain. Later modification is detectable, but the chain does not authenticate

TABLE 3
Evidence regimes, implemented sensors, and demonstrated blind regions.

Regime	Evidence	Implemented sensor family	Exact/empirical blind region	Minimum closing evidence in this study
$\mathcal{I}_X$	Features	Feature JSD, MMD, KS rejection	Every label-only change is structurally blind	Trusted label or item provenance
$\mathcal{I}_{XF}$	Features, scores, predictions	$\mathcal{I}_X$ plus score JSD and prediction disagreement	Every label-only change is structurally blind for a fixed predictor	Item-aligned outcome-prediction evidence
$\mathcal{I}_{Ym}$	Label class counts/marginal	Prior delta and label JSD	Prior-preserving item-level relabeling is structurally blind	Trusted item-level label provenance
$\mathcal{I}_{XFY}$	Item-aligned features, predictions, outcomes	Prediction-rate, disagreement, prediction-distribution, confusion-profile deltas	Closes every positive label-path case executed, conditional on trusted outcomes	Independent authentication of the joint evidence
$\mathcal{I}_Q$	Circuit, parameters, kernel, estimation evidence	Hashes, semantic probes, algebra, repeated estimation, output changes	Sensor-dependent: hashes overreact to equivalence; algebra misses PSD-preserving substitution; outputs miss sub-decision changes	Layered policy-aware composition

TABLE 4
Quantum-gate sensor response fractions.

Condition	Circuit prov.	Kernel prov.	Semantic kernel	Algebraic kernel	Repeated estimate	Output change
Clean	0.00	0.00	0.00	0.00	0.00	0.00
Benign transpilation	1.00	1.00	0.00	0.00	0.00	0.00
Common-unitary rewrite	1.00	1.00	0.00	0.00	0.00	0.00
Parameterized RZ 0.02	1.00	1.00	1.00	0.00	0.00	0.00
Parameterized RZ 0.10	1.00	1.00	1.00	0.00	0.00	0.13
Feature-map reps 1→2	1.00	1.00	1.00	0.00	0.00	0.73
Asymmetric kernel edit	0.00	1.00	1.00	1.00	0.00	0.00
Kernel diagonal erosion	0.00	1.00	1.00	1.00	0.00	0.00
PSD-preserving kernel mix	0.00	1.00	1.00	0.00	0.00	0.07
Shot estimator 256	0.00	1.00	1.00	1.00	1.00	0.47
Shot estimator 1,024	0.00	1.00	1.00	1.00	1.00	0.20

a provider because it is not digitally signed. The response lattice is deliberately fail-closed: all passes yield `allow`; at least one review and no failure yields `hold`; any failure yields `block`. Six prespecified scenarios and all six chains pass eight strict acceptance checks in the frozen artifact.

8 PREREGISTERED REINFORCEMENT GATES (ARTIFACT 1.1.0)

Three sensitivity gates were added after the 1.0.0 evidence was frozen. Their protocol, endpoints, and one implementation clarification (amendment A1) were recorded in `manuscript/paper15_v11_reinforcement_prereg.md` before execution; results are reported as executed. The gates reuse the exact- statevector engine, the frozen split and model seeds, and the frozen 18-condition suite. They add no QPU, calibrated-noise, scheduling, provider, multi-tenancy, or Fleet Management evidence, and they do not change any 1.0.0 count.

8.1 Gate N: Null Calibration with Disjoint Clean Pools

For each of the 240 expansion runs (eight environments, five split seeds, two model seeds, three dimensions), the clean pool is the set of evaluation-side rows not selected into the frozen evaluation batch (900 – n in ID protocols, 3,000 – n in

OOD protocols), transformed with the same training-fitted projection and scaler as the frozen run. A permutation seeded by the split seed divides the pool into a calibration half and an evaluation half. Twenty clean batches of n rows are drawn from each half by simple random sampling without replacement; the suite also includes an identity control and two near-null controls (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$) applied in place.

Ten distributional sensors are calibrated: feature JSD, MMD, and KS rejection rate against the clean batch; score JSD; prediction-rate shift and prediction JSD; label prior shift and label JSD; and the two confusion-profile statistics. For each (environment, dimension, model) cell the 200 calibration values are pooled and the threshold is the 191st order statistic ($\lceil 201 \times 0.95 \rceil$, $\alpha = 0.05$); a sensor fires when its value strictly exceeds the threshold. Thresholds are frozen before any attacked row is examined. The empirical false-alarm rate is the firing fraction on the 200 disjoint evaluation draws of the same cell. A regime fires when any of its sensors fires; regime-level false-alarm rates are therefore unions and are reported as such, without correction.

Two auditors are distinguished (amendment A1). The *batch-level auditor* receives a new batch without item correspondence and can use only the calibrated distributional sensors. The *item-aligned auditor* compares the same items against

TABLE 5

Gate N: empirical false-alarm rate of the null-calibrated sensors and batch-level regimes on 12,000 disjoint clean evaluation draws (all eight environments pooled; nominal $\alpha = 0.05$ per sensor; Clopper–Pearson 95% intervals). Regime rows are unions without correction.

Sensor / regime	Fires	Rate	95% CI
Feature JSD	600	0.050	[0.046, 0.054]
Feature MMD	822	0.069	[0.064, 0.073]
KS rejection rate	342	0.029	[0.026, 0.032]
Score JSD	668	0.056	[0.052, 0.060]
Prediction-rate shift	520	0.043	[0.040, 0.047]
Prediction JSD	586	0.049	[0.045, 0.053]
Label prior shift	570	0.048	[0.044, 0.051]
Label JSD	570	0.048	[0.044, 0.051]
Confusion profile L_1	517	0.043	[0.040, 0.047]
Confusion profile JSD	638	0.053	[0.049, 0.057]
$\mathcal{I}_X$ (any sensor)	1500	0.125	[0.119, 0.131]
$\mathcal{I}_{XF}$ (any sensor)	2438	0.203	[0.196, 0.210]
$\mathcal{I}_{Y_m}$ (any sensor)	570	0.048	[0.044, 0.051]
$\mathcal{I}_{XFY}$ (any sensor)	3113	0.259	[0.252, 0.267]

trusted references, so prediction disagreement, confusion-profile deltas, and item-level label mismatch have an exactly zero null and any non-zero delta is a detection. Item-aligned sensors are undefined on clean draws, whose items differ from the reference batch, and are excluded from batch-level unions. Detection rates apply the frozen thresholds to the frozen `paper_core` observations of the same cell.

Two disclosed limitations are fixed in advance. In E1 ($n = 256$) each pool half holds 322 rows, so successive draws overlap heavily and E1 calibrated rates are indicative only. Simple random draws let class counts vary, whereas the frozen batch is stratified, so the label-marginal null reflects natural batch variability.

Results. All 240 runs completed (14/14 acceptance checks). Table 5 reports the pooled false-alarm rates: every sensor lies between 0.029 and 0.069 against the nominal 0.05, and the per-cell rates have median 0.04 with 92.8% of non-E1 cells inside $[0.01, 0.10]$. Table 6 shows the environment dependence; E1 and E3 (ToN-IoT, 25.7% near-duplicate rows) have the highest regime unions. Table 7 gives the calibrated detection rates and the near-null control responses; Table 8 isolates the label path: the calibrated $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and (prior-preserving) $\mathcal{I}_{Y_m}$ regimes fire on none of the 3,600 label observations, the batch-level $\mathcal{I}_{XFY}$ union fires on 43 (40 of the 2,184 positive impacts), and the item-aligned auditor detects all 3,600. Table 9 records the cluster structure that makes the KS sensor react to in-place perturbations of order 10^{-3} standard deviations while its between-batch null stays narrow; this fingerprint is legitimate for unmodified data but would be evaded by perturbations that preserve the clusters.

8.2 Gate P: Symmetric Preprocessing Ablation

The CICIDS ID design (128/128, `paper_core`, ZZ and SVC, five splits, two model seeds, three dimensions) is re-executed under three scaler configurations: P0, the frozen reference (standard for SVC, $\text{minmax}2\pi$ for QSVC); PA, standard for both; PB, $\text{minmax}2\pi$ for both. The endpoint is the within-split ZZ-minus-SVC difference in suite-average balanced-accuracy conclusion impact with a 95% t interval over the five split clusters; clean balanced accuracy and the exact label-path invariance counts are reported for every configuration.

8.3 Gate T: Cross-Validated Tuning of Both Learners

The same design is re-executed with both learners tuned on training rows only: SVC over $C \in \{0.1, 1, 10, 100\} \times \gamma \in \{\text{scale}, 0.01, 0.1, 1\}$ and QSVC over $C \in \{0.1, 1, 10, 100\}$ on the precomputed training fidelity kernel, with stratified five-fold cross-validation (shuffle, random state = model seed), balanced-accuracy scoring, and ties resolved toward the smallest C . Two environments were fixed before execution: CICIDS ID and the expansion OOD environment with the highest mean clean SVC balanced accuracy in the frozen 1.0.0 tables, which is UNSW temporal OOD (E8, 0.753; the CICIDS temporal pairs lie between 0.485 and 0.493). Untuned references with identical seeds are P0 for CICIDS ID and the frozen E8 rows for UNSW OOD.

Results of Gates P and T. Table 10 reports the paired differences and Table 11 the clean balanced accuracies and selected regularisation constants. The direction of the secondary profile is stable (five of five positive split clusters in every configuration, dimension and environment), but its magnitude is conditional: standardizing the quantum branch (PA) reduces the difference to roughly a third of the reference, matched min-max scaling (PB) leaves it unchanged, tuning both learners widens it in CICIDS ID as ZZ gains clean headroom, and leaves UNSW temporal OOD unchanged while training-only tuning slightly lowers the OOD accuracy of SVC. Exact label-path invariance holds in every configuration (360/360 label rows blind to feature/prediction evidence, 180/180 prior-preserving rows blind to label marginals, every positive impact with joint response, no negative impact).

9 REPRODUCTION AND EVIDENCE MANIFEST

The recommended read-only verification is:

```
python -m src.experiments.\
verify_publication_artifact \
--root publication/artifact
```

It verifies the artifact-wide SHA-256 manifest, five embedded evidence manifests, 44 outputs, row counts, primary-claim counts, and the calibrated label-path consistency checks of the reinforcement gate. In artifact 1.1.0 it reports 3,600 expansion label rows, 2,184 positive expansion impacts, 1,440 Gate-1 label rows, 1,276 positive Gate-1 impacts, and 60 calibrated (environment, dimension, model) cells. The 26-test suite covers invariances, the exact kernel, the contracts, the null-control sampler, the frozen configuration fingerprint, the prespecified threshold rank, and the cross-validation tie rule.

The complete test command is:

```
python -m pytest tests -q -p no:cacheprovider
```

The frozen environment uses Python 3.10.16, NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, Qiskit 2.3.0, and Qiskit Machine Learning 0.9.0. Qiskit is pinned below 3 because migrating deprecated feature-map constructors would change circuit serialization and therefore requires a new evidence version.

The artifact intentionally omits raw benchmark datasets. It includes staging instructions, source locations, source code, locked dependencies, compact derived evidence, figures, figure-source tables, tests, manifests, and expected outputs. Public DOI insertion remains contingent on confirmed authorship, license, and repository visibility; those administrative fields do not alter the scientific results.

TABLE 6

Gate N: per-environment false-alarm rates on 1,200 evaluation draws (three dimensions, two or four models, 200 draws per cell). Sensor range is the minimum and maximum per-sensor rate in the environment; regime columns are batch-level unions. E1 draws 256 rows from 322-row pool halves and is indicative only.

Environment	n	Pool half	Sensor range	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
E1 CICIDS ID 256	256	322	0.023–0.130	0.203	0.245	0.045	0.331
E2 UNSW-NB15 ID	128	386	0.008–0.070	0.063	0.135	0.070	0.210
E3 ToN-IoT ID	128	386	0.025–0.172	0.215	0.280	0.070	0.358
E4 Tue→Wed	128	1436	0.025–0.082	0.150	0.230	0.025	0.254
E5 Tue→Fri-PortScan	128	1436	0.018–0.051	0.066	0.160	0.025	0.198
E6 Wed→Thu-Web	128	1436	0.013–0.057	0.070	0.163	0.025	0.181
E7 Wed→Fri-AM	128	1436	0.025–0.066	0.133	0.212	0.025	0.258
E8 UNSW temporal OOD	128	1436	0.032–0.092	0.072	0.192	0.050	0.237

TABLE 7

Gate N: detection rate of each regime on the frozen `paper_core` observations (all environments, dimensions and models pooled; 600 observations per intervention row) at the null-calibrated thresholds, and firing rate on the in-place near-null controls. Batch-level regimes use calibrated distributional sensors; item-aligned regimes use exact zero thresholds on item-wise deltas.

Intervention	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	$\mathcal{I}_{XF}$ item	$\mathcal{I}_{XFY}$ item
Feature sign flip 0.02	0.118	0.192	0.000	0.200	0.793	0.793
Feature sign flip 0.05	0.532	0.640	0.000	0.643	0.867	0.867
Feature sign flip 0.10	0.757	0.877	0.000	0.878	0.907	0.907
Mean shift 0.02	1.000	1.000	0.000	1.000	0.357	0.357
Mean shift 0.05	1.000	1.000	0.000	1.000	0.500	0.500
Mean shift 0.10	1.000	1.000	0.000	1.000	0.618	0.618
Scaling drift 0.02	1.000	1.000	0.000	1.000	0.537	0.537
Scaling drift 0.05	1.000	1.000	0.000	1.000	0.608	0.608
Scaling drift 0.10	1.000	1.000	0.000	1.000	0.668	0.668
Feature dropout 0.02	0.000	0.000	0.000	0.000	0.488	0.488
Feature dropout 0.05	0.000	0.022	0.000	0.022	0.705	0.705
Feature dropout 0.10	0.000	0.062	0.000	0.062	0.763	0.763
Prior-preserving label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Prior-preserving label flip 0.05	0.000	0.000	0.000	0.002	0.000	1.000
Prior-preserving label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Random label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Random label flip 0.05	0.000	0.000	0.000	0.003	0.000	1.000
Random label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Near-null Gaussian $\sigma = 0.001$	0.713	0.713	0.000	0.713	0.042	0.042
Near-null scaling $\alpha = 0.001$	0.879	0.882	0.000	0.882	0.099	0.099

TABLE 8

Gate N: label-path interventions in the frozen expansion; number of observations on which each regime fires after calibration (batch-level) or by exact item-aligned comparison.

Subset	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	item-aligned
All label interventions	3,600	0	0	0	43	3,600
Prior-preserving	1,800	0	0	0	21	1,800
Positive conclusion impact	2,184	0	0	0	40	2,184

10 CLAIMS-SUPPORTED / CLAIMS-EXCLUDED MATRIX

Supported by the artifact. Auditability is conditional on evidence exposed at a declared boundary; model-output and evaluation-conclusion impacts can have different causal loci; feature/prediction sensors are exactly blind to label-only changes, whereas label marginals are blind to prior-preserving relabeling; these blind regions persist under null calibration at $\alpha = 0.05$ and are closed only by item-aligned evidence; non-invariant sensors reach nominal false-alarm rates on disjoint clean pools within the frozen design; the secondary ZZ-minus-SVC profile is direction-stable but scaler- and headroom-dependent; complementary quantum-

TABLE 9

Cluster structure of the frozen evaluation batches (mean over the 30 cells of each environment, standardized projected features): fraction of near-duplicate rows, and the largest fraction of rows within a window of 0.01 or 0.001 standard deviations in any feature (max) or on average over features (mean).

Environment	Dup. rows	Max 0.01	Mean 0.01	Max 0.001
E1 CICIDS ID 256	0.074	0.825	0.562	0.648
E2 UNSW-NB15 ID	0.114	0.735	0.458	0.514
E3 ToN-IoT ID	0.257	0.870	0.466	0.571
E4 Tue→Wed	0.054	0.766	0.489	0.480
E5 Tue→Fri-PortScan	0.129	0.872	0.747	0.822
E6 Wed→Thu-Web	0.004	0.827	0.433	0.696
E7 Wed→Fri-AM	0.026	0.769	0.670	0.663
E8 UNSW temporal OOD	0.100	0.694	0.437	0.477

workflow sensors close different simulator-gate blind regions; and the local prototype executes policy-aware, fail-closed allow/hold/block composition.

Explicitly excluded. The evidence does not estimate attack prevalence or population-level deployment incidence, establish universal superiority or vulnerability of quantum kernels, or causally isolate feature-map geometry from preprocessing and clean headroom. It provides no evidence

TABLE 10

Gates P and T: within-split ZZ-minus-SVC difference in suite-average balanced-accuracy conclusion impact, mean with 95% t interval over five split clusters and number of positive clusters. CICIDS ID reference is the re-executed P0 configuration; UNSW OOD reference is the frozen 1.0.0 environment E8.

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2 π)	0.0245 [0.0069, 0.0422] (5/5)	0.0348 [0.0176, 0.0520] (5/5)	0.0407 [0.0305, 0.0509] (5/5)
PA standard / standard	0.0070 [0.0014, 0.0125] (5/5)	0.0130 [0.0041, 0.0219] (5/5)	0.0111 [0.0052, 0.0170] (5/5)
PB minmax2 π / minmax2 π	0.0238 [0.0054, 0.0423] (5/5)	0.0323 [0.0156, 0.0491] (5/5)	0.0396 [0.0261, 0.0531] (5/5)
CICIDS ID, both tuned (CV5)	0.0426 [0.0094, 0.0757] (5/5)	0.0713 [0.0494, 0.0932] (5/5)	0.0776 [0.0610, 0.0943] (5/5)
UNSW OOD, frozen defaults	0.0226 [0.0148, 0.0304] (5/5)	0.0379 [0.0260, 0.0498] (5/5)	0.0483 [0.0400, 0.0565] (5/5)
UNSW OOD, both tuned (CV5)	0.0229 [0.0103, 0.0356] (5/5)	0.0348 [0.0205, 0.0492] (5/5)	0.0459 [0.0371, 0.0546] (5/5)

TABLE 11

Gates P and T: clean balanced accuracy (SVC / ZZ, mean over five split clusters) per configuration and dimension, and the distribution of cross-validated regularisation constants over the 30 cells of each tuned environment (SVC also selects γ ; see the artifact table).

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2 π)	0.753 / 0.734	0.747 / 0.728	0.778 / 0.747
PA standard / standard	0.753 / 0.727	0.747 / 0.720	0.778 / 0.727
PB minmax2 π / minmax2 π	0.751 / 0.734	0.750 / 0.728	0.756 / 0.747
CICIDS ID, both tuned (CV5)	0.741 / 0.750	0.746 / 0.781	0.780 / 0.802
UNSW OOD, frozen defaults	0.748 / 0.760	0.754 / 0.766	0.755 / 0.769
UNSW OOD, both tuned (CV5)	0.739 / 0.759	0.723 / 0.757	0.736 / 0.770
Tuned environment	QSVC C selections		SVC C selections
CICIDS ID	$C = 0.1: 3, C = 1: 4, C = 10: 6, C = 100: 17$		$C = 0.1: 1, C = 1: 6, C = 10: 15, C = 100: 8$
UNSW OOD	$C = 1: 17, C = 10: 13$		$C = 0.1: 1, C = 1: 2, C = 10: 15, C = 100: 12$

about QPU security, calibrated device noise, scheduling, provider integrity, multi-tenancy, side channels, production readiness, provider authentication, non-repudiation, persistence, incident handling, or operational Fleet Management.